

Shrey Tiwari

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My research focuses on building practical tools to improve the reliability of industry-scale software by studying complex, understudied bugs and developing static and dynamic analyses to detect them.

Education

Carnegie Mellon University - School of Computer Science

Aug 2023 - Jul 2028 (expected)

Doctor of Philosophy, Advisor: Prof. Rohan Padhye

CGPA: **4.0/4.0**

PES University

Aug 2016 - Jul 2020

B.Tech (honors) in Computer Science & Engineering (Systems & Core Computing), **Gold Medalist**

CGPA: **9.93/10.0**

Research

Chronically Buggy: Analyzing Date/Time Pitfalls In Open-Source And LLM-Generated Python Code

EMSE 2025

Shrey Tiwari, Alexander Joukov, Peter Vandervelde, Serena Chen, Leyi Cui, Ao Li, Rohan Padhye

Under Review

It's About Time: An Empirical Study of Date and Time Bugs in Open-Source Python Software

MSR 2025

Shrey Tiwari, Alexander Joukov, Peter Vandervelde, Serena Chen, Ao Li, Rohan Padhye

🏆 ACM SIGSOFT Distinguished Paper Award

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM

OOPSLA 2025

Ao Li, Byeongjee Kang, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye

Lightweight and Modular Resource Leak Checking (Extended Version)

SOAP 2024

Pritam Gharat, Narges Shadab, Shrey Tiwari, Michael D. Ernst, Shuvendu Lahiri, Akash Lal, Manu Sridharan

Inference of Resource Management Specifications

OOPSLA 2023

Pritam Gharat, Narges Shadab, Shrey Tiwari, Michael D. Ernst, Shuvendu Lahiri, Akash Lal, Manu Sridharan

Resource Leak Checker (RLC#) for C# code using CodeQL

Preprint 2023

Narges Shadab, Pritam Gharat, Shrey Tiwari, Shuvendu Lahiri, Akash Lal

Work Experience

Uber Technologies, Inc.

Jun 2025 - Aug 2025

Research Intern | Advisors: Dr. Sonal Mahajan & Dr. Yuxin Wang

California, USA

- Worked with the Programming Systems group to improve software quality in production by improving automated program analysis.
- Enhanced the accuracy of automated code reviews in uReview and improved the precision of the static nil-panic detector, NilAway.

Microsoft Research

Jun 2021 - Jul 2023

Research Fellow | Advisors: Dr. Akash Lal & Dr. Suman Nath

Bangalore, India

- At Microsoft Research India's Cloud Reliability team, I developed tools to "shift left" Azure reliability and prevent production bugs. My research focused on developing static and dynamic resource leak detection tools for the Microsoft Azure cloud.
- I was also exposed to multiple nascent projects utilizing LLMs to reason about code semantics, including bug detection in Python, fixing compilation errors in Rust, and inferring program specifications for static analysis in C#.

Citrix Systems

Jul 2020 - Jun 2021

Software Engineer | Supervisor: Mukul Agarwal [✉](#)

Bangalore, India

- I was the core developer for the Citrix Cloud Connector [✉](#), a critical component of the cloud VPN gateway service, handling over 3 million weekly VPN tunnels.
- I was responsible for feature development, bug fixes, production monitoring, and on-call support. I contributed to increasing the test coverage to >90%, resulting in the Cloud Connector becoming the most reliable component of the cloud VPN solution.

Citrix Systems

Jan 2020 - Jun 2020

Software Development Intern | Supervisor: Mukul Agarwal [✉](#)

Bangalore, India

- I developed a test automation platform for the Citrix VPN Gateway client, reducing testing time from 2 days to 5 hours and eliminating manual testing efforts. I also resolved customer issues and bugs, gaining extensive experience in Windows debugging, including dump analysis and performance monitoring.

Morgan Stanley

May 2019 - Jul 2019

Technology Analyst Intern | Supervisor: Vignesh Natarajan [✉](#)

Bangalore, India

- I containerized the "Data Influx Adapter", a key component of the tax lot processing pipeline, using Docker. Testing on a 6-million tax lot dataset demonstrated significant cost savings while maintaining performance. I successfully deployed the containerized solution to production, improving code portability and developer productivity.

Indian Institute of Science

May 2018 - Aug 2018

Research Intern | Advisor: Prof. Arkaprava Basu [✉](#)

Bangalore, India

- I was selected as one of 19 students (from 2500+ applicants) for the Narendra Summer Internship [✉](#) at IISc's Computer Systems Lab [✉](#), researching dynamic detection of heterogeneous data races in CUDA-based GPGPU programs.

Selected Projects

DateSMT: SMT-Based Reasoning for Calendrical Arithmetic

Sept 2025 - Present

Carnegie Mellon University

- Building DateSMT, the first framework to encode real-world date and period semantics into SMT solvers, unlocking more precise and scalable analyses for tools that rely on symbolic execution, model checking, and formal verification.

AI-Assisted Differential Fuzz Testing of Date and Time Libraries

Mar 2025 - Present

Carnegie Mellon University

- Developing a differential fuzz testing framework that utilizes LLMs to synthesize and translate date/time computations across libraries, aimed at detecting edge-case bugs and ensuring the correctness of Python date/time libraries.

LLMs for Code Generation in No-Resource Languages

Jan 2024 - May 2024

Carnegie Mellon University

- Investigated the application of state-of-the-art LLMs for code generation in "no-resource" programming languages, implementing prompting techniques that increased compile@10 performance to 50% for languages like Onyx and Catala with GPT-4 Turbo.
- Proposed a new programming language design, Dagger, optimized for neural code generation, to address the limitations observed in the Catala language for context-limited code generation tasks.

FixIt: A Precise and Scalable Dynamic Resource-Leak Detection Tool

Aug 2021 - Jul 2022

Microsoft Research

- Developed a novel and scalable lightweight instrumentation-based tool for detecting resource leaks in managed programming languages like C# and Java, achieving approximately 79% accuracy by leveraging unit test suites.

Futuristic Homes: A Practical Take on Secure, Economical and Voice-enabled Smart Home

Aug 2019 - Jun 2020

Automation Systems [✉](#)

Center for Information Security, Forensics and Cyber Resilience (ISFCR), PES University

- Developed a secure, scalable, and platform-agnostic smart home automation solution. Integrated various technologies to create an elegant, robust system for automating and controlling IoT networks in homes and offices.

Mentoring

Research Study	Mentoring an undergraduate student on her final year research project at the PASTA Lab, where she collaborates with me on my ongoing research.	<i>Dec 2024 - Apr 2025</i>
REUSE Program	Mentored three outstanding undergraduate students during the summer, contributing to CMU's recognition as having the best REU mentors in the United States.	<i>Jun 2024 - Aug 2024</i>

Service

Steering Committee Member	Software Engineering Seminar at the Software and Societal Systems department (S3D)	<i>May 2025</i>
Reviewer	Programming Language Design and Implementation conference (PLDI 2025)	<i>Dec 2024</i>
Student Volunteer	International Conference on Systems, Programming, Languages, and Applications: Software for Humanity (SPLASH 2024)	<i>Oct 2024</i>
Reviewer	International Symposium on Software Testing and Analysis (ISSTA 2024)	<i>May 2024</i>

Presentations

Carnegie Mellon University	Invited to deliver a guest lecture in the Program Analysis course (17-819 ↗)	<i>Oct 2025</i>
Microsoft Research	Gave a talk about my research at the Cloud Reliability Workshop	<i>Feb 2022</i>
Morgan Stanley	Presented my work and analysis results to the entire India team (end of internship expo)	<i>Jul 2019</i>
IISc Bangalore	Presented a poster on my research work at the end of internship exhibit	<i>Jul 2018</i>

Awards & Scholarships

Hima and Jive Fellowship	Recognized for research focusing on improving the reliability of date/time computations in software ↗
Distinguished Paper Award	Awarded the ACM SIGSOFT Distinguished Paper Award at MSR 2025 in Ottawa
SIGPLAN Travel Support	Granted travel support to attend SPLASH 2024 in Pasadena
Gold Medalist	Awarded a gold medal for ranking second in the entire batch of 500+ CSE students, PES University
CNR Rao Scholarship	Six-time recipient of the CNR Rao Merit Scholarship (every applicable semester)
MHRD Scholarship	Six-time recipient of the MHRD Merit Scholarship (every applicable semester)

Achievements

MSR Jigsaw Hackathon	Winner of the Microsoft Research hackathon held for all the research fellows and interns, 2021
Epsilon CTF Competition	Winner of the capture the flag competition held as part of the college technical fest, 2018